Natural Language Processing Based on a Text Graph Convolutional Network

Natural Computing and Machine Learning Laboratory (LCoN),
Mackenzie Presbyterian University, São Paulo, (Brazil)

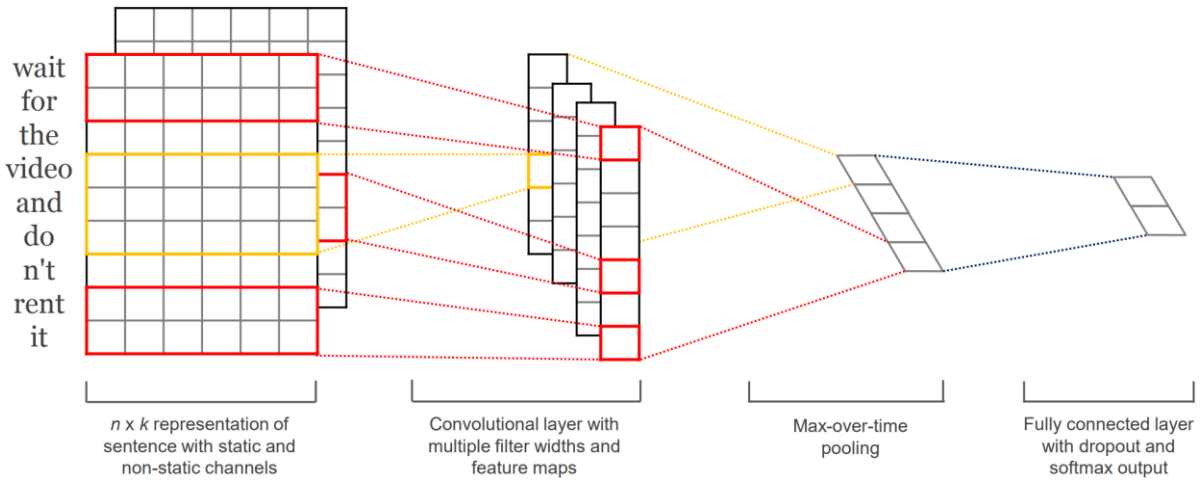
International Conference
DCAI
19th International Conference on Distributed Computing and Artificial Intelligence

Vitor C. M. Pereira
vitormeriat@gmail.com

Leandro Nunes de Castro
lnunes1974@gmail.com

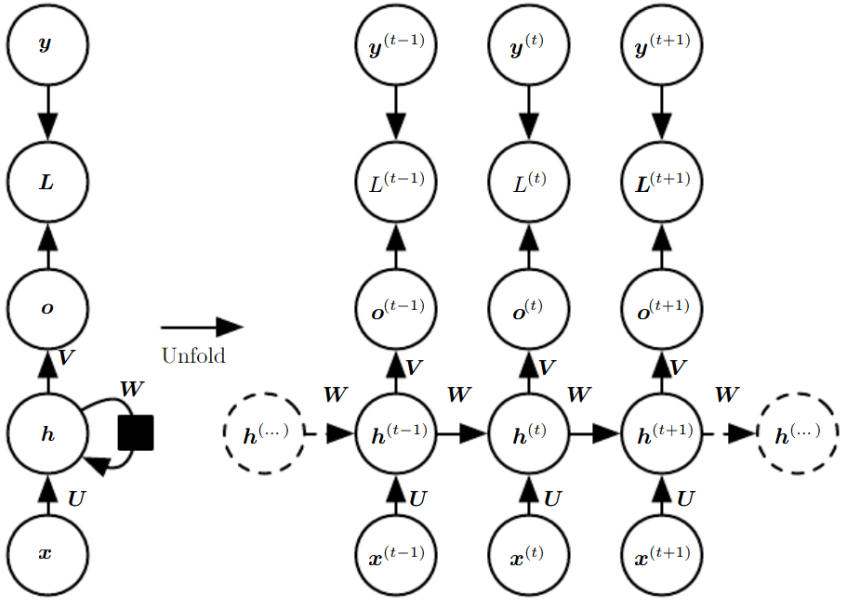
Neural Networks Approaches

CNN



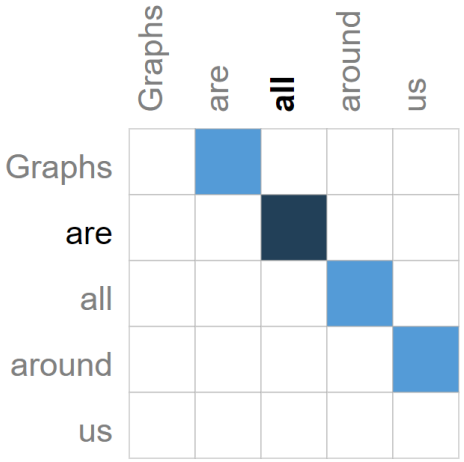
Zhang, Y. & Wallace, B. **A sensitivity analysis of (and practitioners' guide to) convolutional neural networks for sentence classification**. arXiv preprint arXiv:1510.03820 (2015)

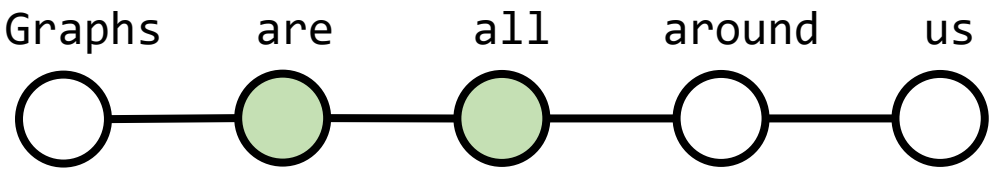
RNN

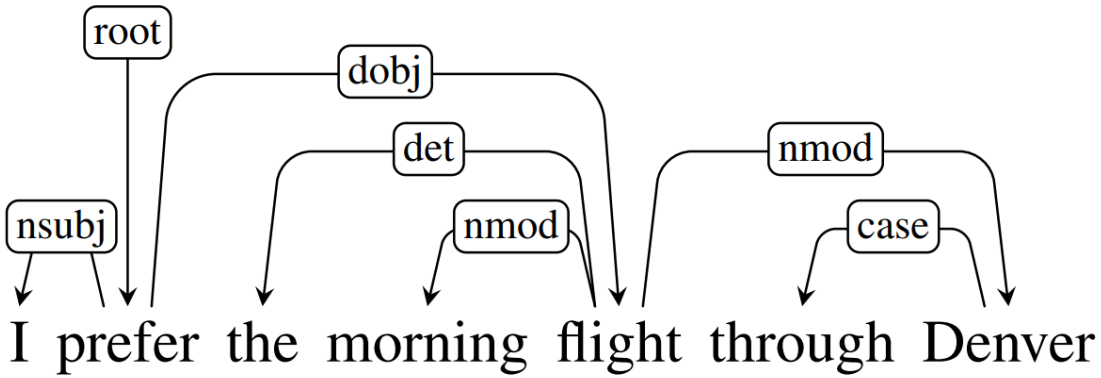


Goodfellow, I., Bengio, Y. & Courville, A. **Deep Learning** (MIT Press, 2016).

Texts as graphs







Using Graph Convolutional Networks (GCN) for Text

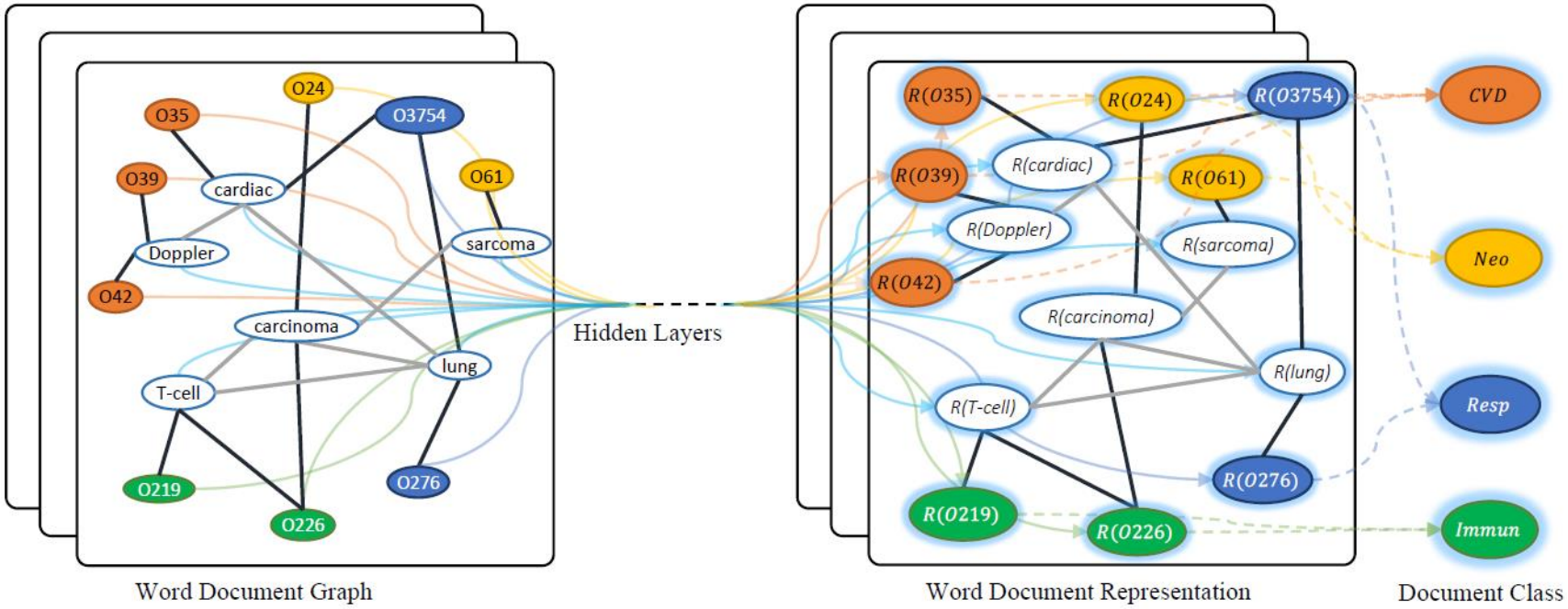
Challenges of Long Text

- CNN and RNN prioritize locality and sequentially.
- They can model local consecutive word sequences well
- They may ignore global word co-occurrence in a corpus

GCN can

- Generalizing well-established neural network models like CNN that apply to regular grid structure (2-d mesh or 1-d sequence) to work on arbitrarily structured graphs
- Can preserve global structure information of a graph in graph embeddings (node, edge, subgraph and whole graph embeddings)

Schematic of Text Graph Convolutional Networks



Yao, L., Mao, C. & Luo, Y. **Graph convolutional networks for text classification**. In Proceedings of the AAAI conference on artificial intelligence, vol. 33, 7370– 7377 (2019)

Building the Text Graph

$$\text{tf-idf}\,(t, d, D) = \text{tf}(t, d) \times \text{idf}(d, D)$$

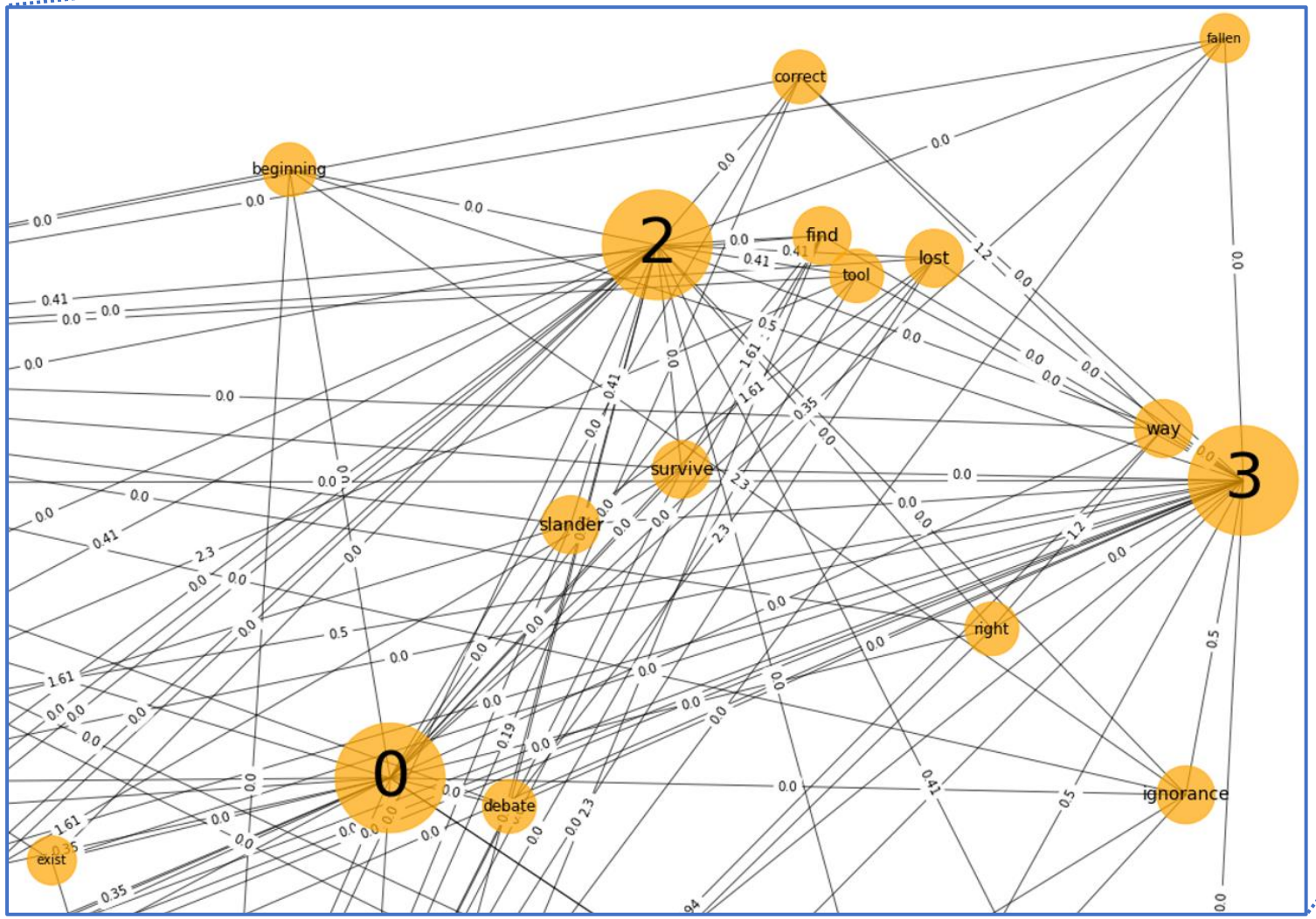
Document content and global word co-occurrence

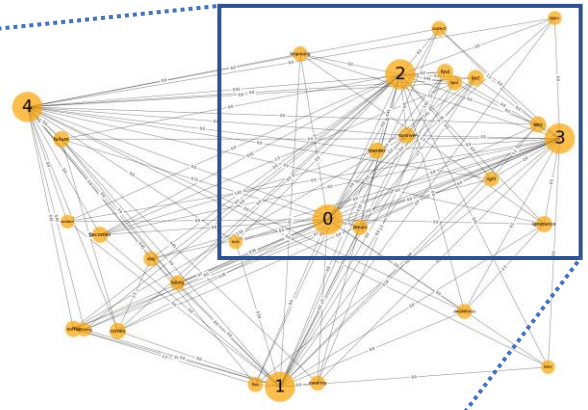
- **Set the number of nodes:** $|V| = d_{size} + v_{size}$
- **Document-word edges:** Term Frequency-Inverse Document Frequency (TF-IDF)
- **Word-word edges:** point-wise mutual information (PMI)

$$\text{PMI}(i, j) = \log \frac{p(i, j)}{p(i)\, p(j)}$$

$$p(i, j) = \frac{\#W(i, j)}{\#W}$$

$$p(i) = \frac{\#W(i)}{\#W}$$


fallen
correct
beginning
2
find
lost
tool
way
3
survive
slander
right
0
debate
ignorance
exist


4
2
3
0
1

Related Word Filter

Document input $doc_i = \{sent_1, sent_2, \cdots, sent_m\}$

Set of word pairs in the sentence $\mathcal{L} = (w_i, w_j)$

$$\mathcal{E}_{ij} = \begin{cases} \text{TF-IDF}(i,j) & i \text{ is the document, } j \text{ is the word} \\ RWF(i,j) & i,j \text{ are words, } RWF(i,j) = 1 \\ 1 & i = j \\ 0 & \text{otherwise} \end{cases}$$

$$\mathcal{E}_{i,j} \left\{ \mathcal{W}_{norm} = \frac{\mathcal{O}(w_i, w_j)}{\mathcal{T}(w_i, w_j)} \right.$$

$$\mathcal{R}_{dep}(i,j) = \begin{cases} 1 & \text{PMI}(i,j) > 0 \\ 0 & \text{PMI}(i,j) \leq 0 \end{cases}$$

Based on the relationships between words provided by the representation



LINGUISTIC

Linguistic Inquiry and Word Count

SYNTACTIC

Dependencies between the words of a sentence based on a grammatical structure

Based on similarity between word vectors



SEMANTIC

Word2Vec with skip-gram or CBOW

A simple two-layer GCN

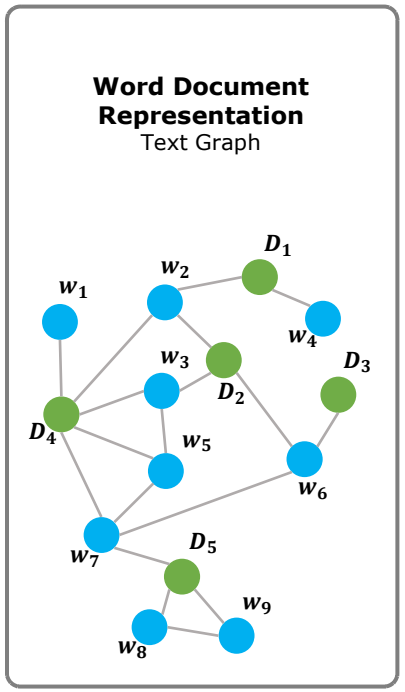
- one-hot feature matrix for words and documents: $X = I$
- 1st layer document and word embeddings: $\tilde{A}XW_0$
- 2nd layer document and word embeddings: $\tilde{A}\,\mathrm{ReLU}(\tilde{A}XW_0)W_1$

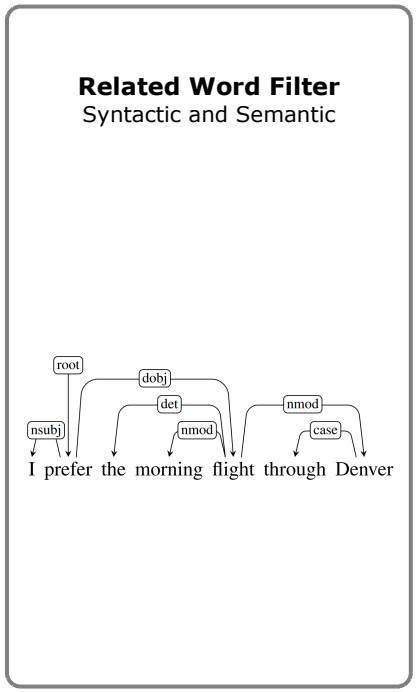
$$Z = \mathrm{softmax}(\tilde{A}\,\mathrm{ReLU}\,(\tilde{A}XW_0)W_1)$$

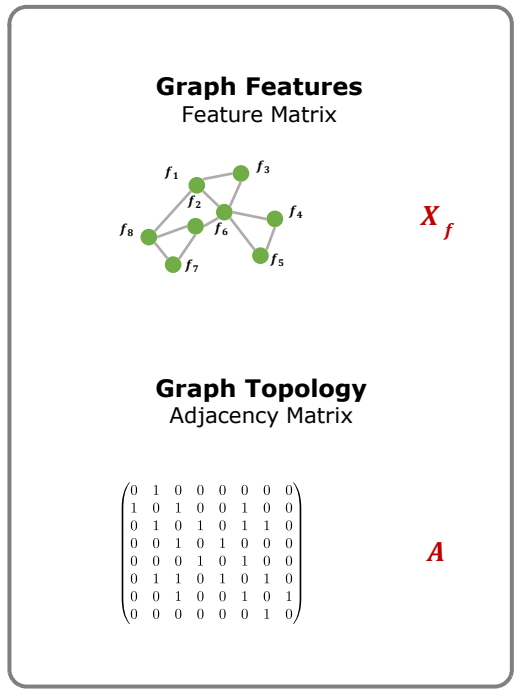
- Loss function: Cross-Entropy Error $\mathcal{L} = -\sum_{d \in \mathcal{Y}_D} \sum_{f=1}^{\mathcal{F}} Y_{df}\, ln\, Z_{df}$

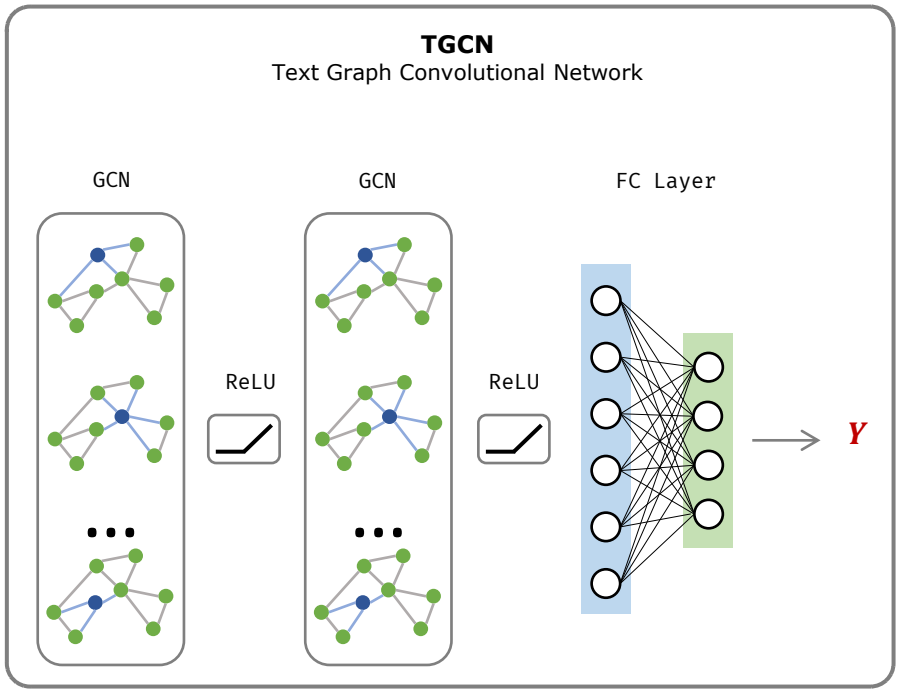
where $\mathcal{Y}_D$ is the set of document índices that have labels and **F** is the dimension of the output features, wich is equal to the number of classes, **Y** is the label indicator matrix

Text Graph Convolutional Network with RWF









Datasets

Dataset	Docs	Words	Nodes	Train	Test	Classes
20NG	18,846	42,757	61,603	11,314	7,532	20
MR	10,662	18,764	29,426	7,108	3,554	2
R8	7,674	7,688	15,362	5,485	2,189	8
R52	9,100	8,892	17,992	6,532	2,568	52
CORA	2,708	1,343	4,051	1,896	812	7
PUBMED	19,717	452	20,169	13,802	5,915	3
CITESEER	3,312	3,515	6,827	2,319	993	6

Dataset split Train&Test: **~70/30**
Dataset Train Validation: **10%**

Results

MODEL	20NG	CITESEER	CORA	MR	PUBMED	R52	R8
TGCN	24,689,966	1,106,622	335,138	1,701,424	929,469	3,981,246	3,165,430
SEMANTIC	2,286,228	173,530	73,336	226,468	849,459	407,694	324,464
SYNTACTIC	3,130,672	105,276	49,344	407,508	850,473	682,675	548,166
LINGUISTIC	2,529,518	104,444	48,918	219,116	850,107	492,946	393,562

MODEL	20NG	CITESEER	CORA	MR	PUBMED	R52	R8
TGCN	0.86126	0.73313	0.75616	0.76027	0.85866	0.93497	0.96848
CNN	0.76931	0.68236	0.69584	0.74981	0.79581	0.85377	0.94037
LSTM	0.65712	0.69281	0.71064	0.75064	0.82984	0.85543	0.93682
SEMANTIC	0.85068	0.71098	0.71921	0.72651	0.86052	0.92445	0.96895
SYNTACTIC	0.84665	0.71702	0.71552	0.73438	0.85123	0.91978	0.95021
LINGUISTIC	0.84241	0.69889	0.72906	0.72313	0.85968	0.91783	0.95432

Conclusions

In this article, we present our proposal for the generation of a new text graph for classification activities based on neural networks for graphs, the Related Word Filter. As the main contribution, we have substantially reduced the size of the text graphic without compromising performance and, in the best case, with a performance superior to that presented by the TGCN.

Future works

- Improving the generation of text graphs that are more representative in terms of context with word embeddings.
- Explore alternatives for processing subgraphs, which would make it feasible to work with large volumes of data.